

## 4.4 From flowers to food

Global food security and sustainability are challenges facing the modern world.

Flowering is a critical stage in the life cycle of a plant and so in the production of staple foods, such as rice and wheat. The timing of flowering must coincide

with favourable environmental conditions so that viable seeds can be produced. Learners will consider the different environmental cues which trigger flowering, the adaptations of flowers to different types of pollination and the formation of seeds and their germination.

### 4.4.1 Plant reproduction

| Learning outcomes                                                                             | Additional guidance                                                                                                                                            |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i> |                                                                                                                                                                |
| (a) the control of flowering in plants                                                        | To include the role of vernalisation and phytochrome in the timing of flowering.<br>HSW2, HSW5, HSW8                                                           |
| (b) adaptations of flowers for pollination                                                    | To include adaptations of both wind-pollinated flowers (e.g. cereals) and insect-pollinated flowers (e.g. legumes).<br><b>PAG2</b><br>HSW4, HSW5, HSW8         |
| (c) fertilisation and seed formation                                                          | To include the growth of the pollen tube, the formation of the embryo and the formation of the endosperm (in outline only).<br><b>PAG2</b><br>HSW4, HSW5, HSW8 |
| (d) (i) the germination of seeds                                                              | To include the role of gibberellin as a cell signalling molecule.<br>HSW1, HSW2, HSW8                                                                          |
| (ii) investigations into the factors affecting germination                                    |                                                                                                                                                                |
| (e) the importance of cereals as staple foods                                                 | To include a consideration of the global use of rice, maize and wheat, and issues relating to food security and sustainability.<br>HSW9, HSW10, HSW12          |